Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B. TECH (ELECTRICAL & ELECTRONICS ENGINEERING)
END SEMESTER EXAMINATIONS, DECEMBER 2024

SOLID STATE DRIVES [ELE 4088]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: DECEMBER 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. An industry which manufactures biodegradable products requires controlled motoring and regenerative braking mode of operation with bidirectional voltage input. Suggest a suitable DC-DC converter circuit for this application. Explain its operation with the help of the converter's output voltage, output current and source current waveforms. **(04)**

1B. Consider a single phase controlled rectifier fed 180V, 10A, 1000 rpm seperately excited DC motor. The motor has an armature resistance of 2Ω. The converter is supplied with a 230V, 50Hz single-phase AC supply. Enough filter inductance is added to ensure continuous conduction.

a. An application requires a steady speed of 600rpm at full load. What should be the firing angle of the converter to attain this?
b. Can the motor develop 15 Nm torque in the motoring mode at the rated speed at the steady state? Justify your answer.

(04)

1C A plastic recycling plant drive system has parts A & B moving at 10m/s and 7m/s respectively. The load force of part A is 10N & part B is 20N. What should be the power developed by the motor rotating at 1000rpm to drive this load at a steady state? Neglect the losses in the conversion systems **(02)**

2A. A 220V 800rpm 8 A separately excited motor has the armature resistance and inductance of 1Ω and 15mH, respectively. The motor is controlled by a single phase fully controlled rectifier at a firing angle of 60 degrees, with an AC source voltage of 230V, 50Hz. Verify whether a continuous armature current is ensured at this firing angle at 70% of the rated speed.

(Expression for armature current: Single-phase controlled rectifier fed separately excited DC motor

$$i_{(\omega t)} = \frac{V_m}{Z}\left[\sin(\omega t - \varphi) - \sin(\alpha - \varphi)e^{((\alpha-\omega t)/\omega\tau)}\right] - \frac{k\omega_m}{R_a}\left[1 - e^{((\alpha-\omega t)/\omega\tau)}\right])$$

(04)

2B. At what firing angle, does a three-phase controlled rectifier-fed DC motor receive an input armature DC voltage of 200V from a 415V three- **(04)**

phase AC supply? Draw the waveforms of armature voltage and current at this condition assuming continuous conduction. Clearly show the triggering sequence. The given template in Fig. Q 2B may be used.

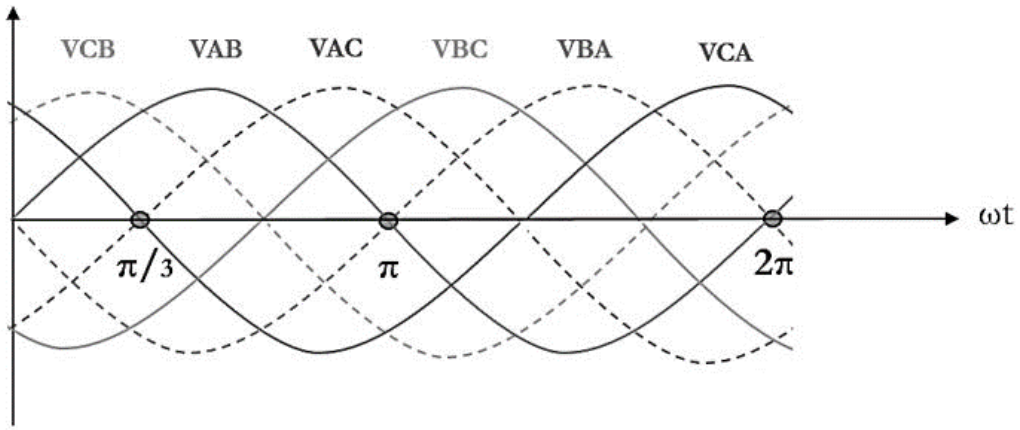


Fig. Q 2B

2C. A cement mixer drive system has to run at 25 rpm driven by a 1500 rpm induction motor. Which control strategy will you suggest out of V/f and flux weakening mode, for the inverter feeding the motor? Justify the reason **(02)**

3A. 'Stator voltage control & rotor resistance control may not be a suitable option for a wide range of speed control of three-phase induction motor'.

Justify this statement with the help of speed-torque characteristics & suitable schematic diagrams of stator voltage control & rotor resistance control of a three-phase induction motor **(03)**

3B. With the help of block schematic discuss the specific applications of different power electronic converters in Electric vehicles. **(04)**

3C. Demonstrate the switching scheme of a circuit with a single IGBT & diode through which how power can be fed back from a separately excited DC motor to the DC source. Draw the waveforms of armature voltage and current **(03)**

4A. Describe how the fictitious d-q coordinates help to realize Field Oriented Control (Vector Control). With a block schematic explain the FOC strategy in any motor. **(04)**

4B. A centrifugal compressor drive system in a petrochemical industry uses a 1500 rpm rated three-phase slip ring induction motor, which needs to be operated in the speed range of 1000 rpm to 2000 rpm. Suggest a control strategy with suitable block schematics of the converter configurations to achieve this speed control where the control is through the electrical circuit of the rotor. Mention the advantages and limitations of the suggested method. **(04)**

4C. Suggest a variable frequency control strategy for constant power AC drives. Draw and explain the block schematic of the suggested strategy. **(02)**

5A. Identify a suitable example of a motor & load in a drive system which shows steady-state stability in the second quadrant of the torque-speed plane. Draw the operating point and demonstrate that the operating point is stable. **(02)**

5B. Permanent Magnet synchronous motors and Brushless DC Motors are popular choices for industrial automation and robotics.
Compare these motors based on construction, working principle and distinct features. **(04)**

5C. List and write an application note on the commonly used EV motors. Also, discuss the emerging EV technologies. **(04)**